

2.5 MW Pulse Power Supply

Team Number: 57

Team Name: Pulse Power Innovators

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Introduction

Pulsed power systems deliver energy in short, high-intensity bursts, enabling very high instantaneous power from modest average sources. They are essential to applications such as particle accelerators, radar, medical devices, and advanced industrial systems. However, full-scale pulsed-power modulators are unsafe and impractical for student laboratories, creating a gap between academic theory and real-world systems.

A low-voltage educational prototype addresses this gap by safely demonstrating pulse formation, energy transfer, and waveform control. This hands-on platform connects theory to measurable behavior while preparing students for careers in high-reliability research and national laboratory environments.

Roadmap

This report first gives a summary of what the team achieved last semester in the Completed Tasks Section. Then, it explains what the goals of this project are via sections like the Project Scope, Budget, and Final Demonstration Day Vision, which explain what the final design is, the budget and bill of materials, and what the Demo Day will consist of. This report then lays out how the team plans on achieving said goals via the Schedule, listing out each task to be done week by week. Finally, the project will conclude by discussing the ABET Considerations.

Completed Tasks

In the previous semester, the team primarily focused on researching pulsed power topologies and creating a circuit schematic that fits the team's goals and budget. There are various different applications in industry that require different parameters in voltage levels or pulse width, which will inherently change the circuit components and connections. The team created a schematic inspired by the single stage pulse generator topology that fits within the scope of the project. More research was conducted to determine the topology is solid state, which resulted in the inclusion of a gate driver that controls the MOSFETs switches. Included in the circuit as well is a STM32 series microcontroller that will manage the pulse width modulation, which must be programmed in an IDE. The team has familiarized themselves with STM32CubeIDE, a software that uses C as the predominant language for program logic, which will be crucial for controlling the embedded systems in the physical build.

Project Scope

This project presents the design, construction, and validation of an adjustable pulsed DC power supply using PWM switching techniques. The hardware features N-channel power MOSFETs, isolated gate drivers, current-sense amplifiers, and microcontroller-based control enabling user-adjustable frequency, duty cycle, voltage, and current limiting. Physical implementation addresses PCB layout for high-frequency switching, thermal management, and EMI mitigation. Experimental validation includes oscilloscope waveform characterization, efficiency analysis, and current-limiting verification, demonstrating practical tradeoffs in switching converter design suitable for motor control, LED driving, and battery charging applications.

Final Demonstration Day Vision

The presentation begins by introducing the project objectives and final specifications (100Hz-10kHz frequency, 1-99% duty cycle, 1-48V output, 0.1-10A current limiting), then provides a hardware walkthrough identifying MOSFETs, gate drivers, current sensor, microcontroller, and explaining the switching topology and PCB layout considerations. The operating demonstration displays waveforms at various frequencies on the oscilloscope, shows voltage output at different duty cycles, demonstrates current limiting under overload, and presents timing measurements. Performance results include efficiency data across the operating range, thermal behavior under sustained operation, and comparison of measured versus theoretical values. A practical application demo shows LED dimming or motor speed control, followed by a brief conclusion summarizing achieved performance and design challenges overcome. The session concludes with Q&A where evaluators ask technical questions about design decisions and circuit behavior.

Budget Report

Total Estimated Project Cost: \$185.00

Category	Cost
Power Components (MOSFETs, gate drivers, capacitors, heatsinks)	\$68.00
Control Electronics (microcontroller, sensors, interface components)	\$42.00
PCB & Assembly Materials (board, terminals, wire, solder)	\$35.00

Connectors & Hardware (jacks, enclosure, mounting) \$18.00

Testing & Documentation (load resistors, thermal paste, printing) \$22.00

Required Lab Equipment (Institution-Provided): Oscilloscope, multimeter, DC power supply (0-50V, 5A), soldering station

In-Progress Tasks

The primary goal for the team this semester is to take the power circuit schematic created the previous semester and physically construct the circuit to be ready for the final demonstration. To achieve this, the team must test the design and ensure the values of important pulse power variables such as voltage, current, pulse width, pulse frequency, and rise time. All of these values can be tested using the Tektronix DPO-4032 oscilloscope provided by the laboratory. Adapting is an important part of the testing process as well, as changes may need to be made if the values are wrong or the circuit does not behave as expected. Another task to be completed is writing the microcontroller program responsible for PWM in the aforementioned STM32CubeIDE, and subsequently testing it to ensure it is functioning properly. The microcontroller and hardware testing must be done almost simultaneously, as physical components are needed to test the MCU's I/O and the hardware needs the MCU to operate correctly. After testing and the desired values are reached, components must be assembled for demonstration day and presented in a manner that is easily understood by the viewer. The team intends to do this by showing oscilloscope output for pulse width / voltage and LED activation during the pulse for visual aid.

Impact of delays

This capstone project is being completed by a hybrid team, consisting of both online and on campus students. In an ideal scenario, software simulations will be done by online students and the hardware part will be built by two on campus students.

However, there is a potential delay, considering shipments or rebuilding of the prototype might take longer than expected in case of failure in the first trials or need for replacing a part. To minimize the risk, our team will make sure to complete all the planning and simulation, much before the Demo day and online orders will be placed accordingly.

Changes to Project Scope

The project scope largely remains the same as last semester, using a lower voltage pulsed power model to demonstrate the behavior of a large scale modulation system. Throughout the design process last semester, the team found the scope to be achievable while challenging which resulted in no major changes to the project objectives.

Schedule

Phase 1: Design & Initial Planning (Week 1, Feb 6 - Feb 12) Finalize circuit schematic, component specifications, and Bill of Materials. Place component orders.
Deliverable: First Written Report (Project Synopsis) - Feb 6

Phase 2: Progress Documentation & Procurement (Weeks 2-3, Feb 13 - Feb 26) Receive and inventory components. Breadboard-test gate driver and current sensing circuits. Deliverable: Video Progress Report - Feb 16

Phase 3: PCB Assembly & Initial Testing (Weeks 4-7, Feb 27 - Mar 26) Solder all components including power stage, control circuitry, and user interface. Perform continuity testing. Power up system and verify voltage rails. Milestone: TA Conference - Mar 23-Apr 1

Phase 4: Firmware Development & Debugging (Weeks 8-9, Mar 27 - Apr 9) Write microcontroller code for PWM generation, user interface, and current limiting. Debug switching behavior and test parameter ranges. Deliverables: Second Written Report (Interim Report) - Apr 7, Capstone Showcase Checklist - Apr 9

Phase 5: Performance Testing & Documentation (Weeks 10-11, Apr 10 - Apr 23) Measure switching waveforms, efficiency, and thermal performance. Test current limiting and capture data. Begin final report and prepare demo setup. Deliverables: Project/Team Assessment - Apr 17, Demo Day Intake Form - Apr 20, Senior Design Prize Submission - Apr 22

Phase 6: Final Demo & Submission (Week 12, Apr 24 - Apr 29) Complete presentation slides and poster. Rehearse demonstration. Finalize technical report with all experimental results. Deliverables: Final Report - Apr 29, Course Evaluation - Apr 29

ABET Considerations

Sustainability: Pulsed power technology is inherently linked to sustainability in the form of its energy usage. The alternative to supplying a quick pulse of voltage is through conventional DC or AC methods, which may not be necessary for all applications. Controlled bursts will **reduce power consumption**, which can be particularly useful in machinery that require extremely high-voltage levels to run, such as a particle accelerator.

Manufacturability: Given the constraints of the supplies allocated to the group, the pulsed power display will be **easier to manufacture** when compared to industrial applications. During the design process, the team is focused on making an affordable demonstration without sacrificing quality of results. This pulse power supply is not meant for high-voltage but to demonstrate to a customer how this method works and how it can be implemented in certain machinery. The team's goal is to design a power supply not only easily comprehensible to a customer, but one that is easily reproducible as well.

Safety: The specific low-voltage topology in this project was chosen with the idea of customer safety in mind. Pulsed power modulation is much more common in high-voltage applications, but due to **safety concerns** commonly present in such situations, a low-voltage model only for demonstration will be used instead. The design will also use devices previously mentioned to ensure safety while testing, such as bleeder resistors and TVS diodes.

Conclusion

Team Pulse Power Innovators has chosen an important project that can prototype a safe high-power supply for the public use. Initial software simulations and complete roadmap for this semester already completed. Hardware parts are being shipped, and the team is close to finalizing the entire project. By the end of the semester, the team intends to showcase a fully-functioning prototype of a Pulse Power Supply.